2D Matrices: Combining Transformations

Questions 1–45

Question 1. The transformations P, Q and R have associated matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ respectively. Write down a product of matrices associated with each combined transformation:

a) P followed by Q
b) R followed by P
c) Q followed by R
d) P followed by R followed by Q
e) Q followed by P followed by R

Question 2. Given matrices $\mathbf{A} = \begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ represent the linear transformations A and B respectively, find:

a) the matrix that represents the transformation A followed by the transformation B
b) the matrix that represents the transformation B followed by the transformation A

Question 3. Find the matrix that represents a reflection in the x-axis followed by a reflection in the y-axis. Hence describe the single transformation represented by this matrix.

Question 4. Given matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, describe fully:

a) the single transformation represented by $\mathbf{A}$
b) the single transformation represented by $\mathbf{A}^{2024}$

Question 5. Given matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, describe fully:

a) the single transformation represented by $\mathbf{A}$
b) the single transformation represented by $\mathbf{A}^{335}$

Question 6. Describe the combination of transformations represented by the following matrices:

a) $\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$
b) $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$
c) $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$
d) $\begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix}$
e) $\begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix}$
f) $\begin{pmatrix} -8 & 0 \\ 0 & 8 \end{pmatrix}$
g) $\begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$

Question 7. It is given that $\mathbf{A} = \begin{pmatrix} 1 & k \\ 3 & 2 \end{pmatrix}$ for constant k, the 2×2 matrix $\mathbf{B}$ has determinant 5 and the transformation represented by the matrix $\mathbf{AB}$ preserves both area and orientation. Find the value of k.

Question 8. It is given that $\mathbf{A} = \begin{pmatrix} 2k+1 & 4 \\ 3k & -2 \end{pmatrix}$ for constant k, the 2×2 matrix $\mathbf{B}$ has determinant -3 and the transformation represented by the matrix $\mathbf{AB}$ preserves area but not orientation. Find the value of k.

Question 9. It is given that $\mathbf{A} = \begin{pmatrix} 4 & p \\ q & p \end{pmatrix}$, where p and q are constants and $\mathbf{A}^2$ represents an enlargement centred at the origin, by scale factor 100. Find the value of p and the value of q.

Question 10. The transformation A is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 5 & 0 \end{pmatrix}$, and is equivalent to a rotation and another transformation.

a) Draw a diagram to show the points $(1, 0)$ and $(0, 1)$ and their images after the transformation A is applied.

b) Fully describe the two transformations that are equivalent to A.

Question 11. Determine the matrix that represents:

a) An enlargement, centred at the origin, by scale factor 4, followed by a stretch, parallel to the y-axis, scale factor 10.

b) A reflection in the x-axis, followed by a reflection in the y-axis.

c) A rotation by 90° anticlockwise about the origin, followed by a stretch, parallel to the x-axis, scale factor 3.

d) A reflection in the line $y = x$, followed by a rotation by 180° about the origin.

e) A rotation by 45° anticlockwise about the origin, followed by a reflection in the line $y = -x$.

f) A reflection in the y-axis, followed by a rotation by 90° clockwise about the origin.

g) A rotation by 90° clockwise about the origin, followed by a reflection in the line $y = x$, followed by a rotation by 90° anticlockwise about the origin.

Question 12. Given $\mathbf{A} = \begin{pmatrix} 5 & -2 \\ 5 & -1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} \frac{4}{15} & -\frac{1}{3} \\ 1 & -1 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ represent linear transformations A, B and C respectively, give a geometrical interpretation of the transformation D, which is found by applying C, then B, and then A.

Question 13. Given $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -4 & 7 \\ 1 & -2 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ represent linear transformations A, B and C respectively, give a geometrical interpretation of the transformation D, which is found by applying B, then A, and then C.

Question 14. The triangle X has vertices at $(\sqrt{2}, 2)$, $(-3, \sqrt{2})$ and $(0, -\sqrt{2})$. X is rotated by 225° anticlockwise about the origin, and then reflected in the line $y = -x$, and finally enlarged by scale factor $\sqrt{2}$ about the origin to form the triangle X'. Using matrices, find the exact coordinates of the vertices of triangle X'.

Question 15. It is given that $\mathbf{A}$ is the 2×2 matrix which represents an enlargement, centred at the origin, by scale factor k, followed by a rotation by $\theta°$, anticlockwise about the origin. It is also given that the transformation associated with $\mathbf{A}$ maps $(2\sqrt{2}, \sqrt{2})$ to its image $(-8, -24)$. Find the values of k and θ.

Question 16. The points $A(1, 2)$ and $B(4, 6)$ are stretched, parallel to the x-axis, by scale factor k, and then stretched, parallel to the y-axis, by scale factor $\frac{1}{k}$, onto their images A' and B', where k is a positive integer. It is given that the distance between A' and B' is $\sqrt{145}$ units. Find the value of k.

Question 17. Given $\mathbf{A} = \begin{pmatrix} 2 & -5 \\ -7 & 4 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix}$ and $\mathbf{C} = 3\mathbf{A} - \mathbf{B}$, and that matrix $\mathbf{C}$ represents a combination of an enlargement and a reflection, in either order, fully describe the two transformations.

Question 18. Given $\mathbf{A} = \begin{pmatrix} 2 & 5 \\ 1 & 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -100 & -40 \\ 40 & 20 \end{pmatrix}$, $\mathbf{C} = \mathbf{AB}$ and that matrix $\mathbf{C}$ represents a combination of an enlargement and a rotation, in either order, fully describe the two transformations.

Question 19. Given matrix $\mathbf{A} = \begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix}$ and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 11, find the value of k and the value of m.

Question 20. Given the matrix $\mathbf{A} = \begin{pmatrix} -5 & 4k \\ -2k & p \end{pmatrix}$, where k and p are positive constants and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 9, find the value of k and the value of p.

Question 21. Given $\mathbf{A} = \begin{pmatrix} 3 & -5 \\ 5 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and that $\mathbf{AB}$ represents a rotation by 90° clockwise about the origin, followed by an enlargement, centred at the origin, by scale factor 12, find the values of a, b, c and d.

Question 22. It is given that $\mathbf{A} = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}$

a) Show that $\mathbf{A}^3 = k\mathbf{I}$, for some constant k.
b) Describe the transformation represented by $\mathbf{A}^3$.
c) Hence describe the transformation represented by $\mathbf{A}$.

Question 23. It is given that $\mathbf{A} = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}$

a) Show that $\mathbf{A}^4 = k\mathbf{I}$.
b) Describe the transformation represented by $\mathbf{A}^4$.
c) Describe the transformation represented by $\mathbf{A}$.
d) It is given that $\mathbf{BA} = \mathbf{I}$. Describe the transformation represented by $\mathbf{B}$.

Question 24. Given $\mathbf{A} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, where a and b are constants.

a) Find $\mathbf{A}^2$ in terms of a and b.
b) It is given that $a = \frac{1+\sqrt{3}}{4}$ and $b = \frac{\sqrt{3}-1}{4}$. Show that $\mathbf{A}^2 = k \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$.
c) Describe the transformation represented by:
   a) $\mathbf{A}^6$
   b) $\mathbf{A}^{12}$
   c) $\mathbf{A}^{24}$
   d) $\mathbf{A}$

Question 25. Given that the transformation T is a reflection in the line $y = \frac{\sqrt{3}}{3}x$, and that the transformation U is a reflection in the line $y = \sqrt{3}x$. Find the matrix that represents the combined transformation:

a) T followed by U

b) U followed by T

Question 26. It is given that $\mathbf{A} = \begin{pmatrix} 0.28 & -0.96 \\ 0.96 & 0.28 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -0.8432 & 0.5376 \\ 0.5376 & 0.8432 \end{pmatrix}$

a) Describe the transformation represented by the matrix $\mathbf{A}$

b) Describe the transformation represented by the matrix $\mathbf{B}$

c) Given that $\mathbf{C} = \mathbf{AB}$ represents a reflection, find the equation of the mirror line.

Question 27. Find the matrix that represents a reflection in the line $y\sqrt{3} - 3x = 0$, followed by a reflection in the line $y\sqrt{3} + x = 0$. Fully describe the single transformation that is represented by this matrix.

Question 28. It is given that $\mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix}$

a) Describe the transformation represented by $\mathbf{A}^2$.

b) Given that $\mathbf{A}$ represents a combination of an enlargement and a reflection, find the equation of the mirror line of the reflection.

Question 29. Given $\mathbf{A} = \begin{pmatrix} -\cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}$ maps the point $(1, -1)$ onto the image $(0, -\sqrt{2})$ for some positive value of θ, find the smallest possible value of θ.

Question 30. Given $\mathbf{A} = \begin{pmatrix} 3\cos\theta & -\sin\theta \\ \sin\theta & 3\cos\theta \end{pmatrix}$, where θ is a positive acute angle, maps the point $(3, -\frac{1}{\sqrt{3}})$ onto the image $(k, 0)$, find the value of k.

Question 31. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \cos 40° & -\sin 40° \\ \sin 40° & \cos 40° \end{pmatrix}$

b) $\begin{pmatrix} \cos 40° & \sin 40° \\ -\sin 40° & \cos 40° \end{pmatrix}$

c) $\begin{pmatrix} \cos 40° & \sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 40° & -\sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

Question 32. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} -\cos 40° & -\sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

b) $\begin{pmatrix} -\cos 40° & \sin 40° \\ -\sin 40° & -\cos 40° \end{pmatrix}$

c) $\begin{pmatrix} \cos 40° & -\sin 40° \\ -\sin 40° & -\cos 40° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 40° & \sin 40° \\ \sin 40° & \cos 40° \end{pmatrix}$

Question 33. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \cos 140° & -\sin 140° \\ \sin 140° & \cos 140° \end{pmatrix}$

b) $\begin{pmatrix} \cos 140° & \sin 140° \\ -\sin 140° & \cos 140° \end{pmatrix}$

c) $\begin{pmatrix} \cos 140° & \sin 140° \\ \sin 140° & -\cos 140° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 140° & \sin 140° \\ \sin 140° & \cos 140° \end{pmatrix}$

Question 34. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \sin 40° & -\cos 40° \\ \cos 40° & \sin 40° \end{pmatrix}$

b) $\begin{pmatrix} \sin 40° & -\cos 40° \\ -\cos 40° & -\sin 40° \end{pmatrix}$

c) $\begin{pmatrix} \sin 40° & \cos 40° \\ -\cos 40° & \sin 40° \end{pmatrix}$

d) $\begin{pmatrix} -\sin 40° & \cos 40° \\ \cos 40° & \sin 40° \end{pmatrix}$

Question 35. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \sin 140° & -\cos 140° \\ \cos 140° & \sin 140° \end{pmatrix}$

b) $\begin{pmatrix} \sin 140° & \cos 140° \\ -\cos 140° & \sin 140° \end{pmatrix}$

c) $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix}$

d) $\begin{pmatrix} -\sin 140° & \cos 140° \\ -\cos 140° & -\sin 140° \end{pmatrix}$

Question 36. Given $\mathbf{A} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, $\mathbf{B}$ is the matrix that represents a rotation by 150°, anticlockwise about the origin, and that $\mathbf{A}^2 = \mathbf{B}$, show that $\cos 75° = \sqrt{\frac{2-\sqrt{3}}{4}}$.

Question 37. Given $\mathbf{A} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, $\mathbf{B}$ is the matrix that represents a rotation by 105°, anticlockwise about the origin, and that $\mathbf{A}^2 = \mathbf{B}$, show that $\sin 52.5° = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$.

Question 38. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. Fully describe both transformations in each of these situations:

a) $\mathbf{A} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix}$

d) $\mathbf{A} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix}$

e) $\mathbf{A} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$

f) $\mathbf{A} = \begin{pmatrix} 12-4\sqrt{3} & 12+4\sqrt{3} \\ -12-4\sqrt{3} & 12-4\sqrt{3} \end{pmatrix}$

Question 39. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. The other transformation is a reflection in the line $y = -x$. Given that $|\mathbf{A}| = -12$, write down the matrix $\mathbf{A}$.

Question 40. The triangle X has vertices at $(-\sqrt{3}, -\sqrt{2})$, $(-4\sqrt{3}, -\sqrt{2})$ and $(-3\sqrt{3}, -6\sqrt{2})$. X is stretched, parallel to the y-axis, by scale factor $2\sqrt{2}$, and then rotated by 30° anticlockwise about the origin, and then stretched, parallel to the x-axis, by scale factor $5\sqrt{3}$. Using matrices, find the exact coordinates of the vertices of triangle X' and the shape's area.

Question 41. Given $\mathbf{A} = \begin{pmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{pmatrix}$ represents a composition of a rotation about the origin represented by the matrix $\mathbf{P}$, a reflection represented by the matrix $\mathbf{Q}$, and an enlargement represented by the matrix $\mathbf{R}$, in that order, show how $\mathbf{A}$ can be written as the product of $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$.

Question 42. The matrix $\mathbf{A}$ represents a shear with the x-axis fixed. It is also given that the matrix $\mathbf{B}$ represents a shear with the y-axis fixed. The matrix $\mathbf{M} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$ is a composition of the matrices $\mathbf{A}$ and $\mathbf{B}$. Find the matrices $\mathbf{A}$ and $\mathbf{B}$.

Question 43. Given $\mathbf{A} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \frac{1}{2}\begin{pmatrix} \sqrt{2} & -6\sqrt{2} \\ \sqrt{2} & -4\sqrt{2} \end{pmatrix}$ represent the transformations A and B respectively,

a) describe the transformation A

It is given that B represents the combination of applying transformation A and then transformation C.

a) Find the matrix C and hence describe the transformation C.

Question 44. It is given that $\mathbf{ABC} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$. A and C both represent shears with the x-axis fixed. B represents an anticlockwise rotation about the origin by $90°$. Find the matrices A, B and C.

Question 45. The matrix **M** represents a composition of a rotation about the origin, followed by a stretch, followed by a shear. The triangle X with vertices at $A(0,2)$, $B(3,0)$ and $C(3,2)$ is mapped to its image X' with vertices at $A'(-4,12)$, $B'(0,3)$ and $C'(-4,15)$. Write the matrix **M** as a product of three matrices, with each matrix representing the rotation, stretch and shear.

